Ratio Analysis

Exam code: 9BS0

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An introduction to ratio analysis

- Ratio analysis involves **extracting information from financial accounts** to assess **business performance** and answer key questions, including:
  - Why is one business more **profitable** than another in the same industry?
  - Is the business **growing**?
  - How effectively is the business **using its assets and capital** invested?
  - What **returns on investment** are expected?
  - How **risky** is the **financial structure** of the business?

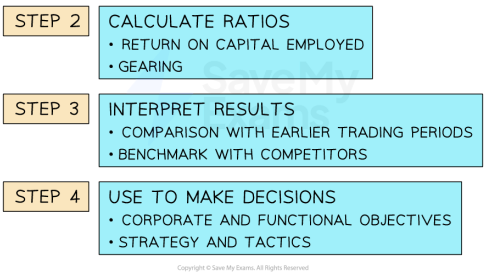
Information extracted from the statement of comprehensive income and the statement of financial position for ratio analysis

Statement of comprehensive income	Statement of financial position
• Revenue • Cost of sales • Gross profit • Operating profit • Profit for the year (net profit)	• Current assets • Current liabilities • Inventory (stock) • Trade receivables • Trade payables • Long-term liabilities • Capital and reserves

- Ratio analysis supports evidence-based decision-making, as it **provides measurable data** that can be used to **support judgements** and **compare performance** against objectives

The ratio analysis process

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it



The ratio analysis process involves collecting data, calculating ratios, interpreting results and then using results to make decisions

Types of ratios

Gearing ratio

- The gearing ratio shows the **long-term financial structure** of the business
  - It shows the balance of **non-current liabilities** (e.g. long-term loans) to **shareholder capital** used to fund a business
  - The outcome is expressed as a **percentage**
- The gearing ratio is calculated using the formula

$$\text{Gearing Ratio} = \frac{\text{Non Current Liabilities}}{\text{Capital Employed}} \times 100$$

- **Capital employed** can be calculated by adding non-current liabilities to total equity

Worked Example

The table shows an extract from the company accounts of *Keals Cosmetics*.

Current Assets	£6.2 million
Current Liabilities	£3.4 million
Non-current Liabilities	£9.6 million
Capital Employed	£43.3 million

Calculate the gearing ratio of *Keals Cosmetics*.

(2)

Step 1: Identify the data required to calculate the gearing ratio

- Non-current liabilities = £9.6 million
- Capital employed = £43.3 million

Help me understand

$$= £45.5 \text{ million} \div £9.6 \text{ million}$$ (1)

$$= 0.22$$

Step 3: Multiply the outcome by 100, and express the result as a percentage

$$= 0.22 \times 100$$ (1)

$$= 22\%$$

- **22%** of Keals Cosmetics' capital structure is made up of long-term loans

Return on capital employed (RoCE)

- The return on capital employed is also known as the **primary ratio**
- It compares the **profit** made by a business to the amount of **capital invested** in the business
- It is a measure of **how effectively a business uses the capital invested** in the business **to generate profit**
- Return on capital employed is a **key performance indicator** that can be **compared over time** and also with **competitors** and with **other potential capital investments**

- Return on capital employed is expressed as a percentage and can be **calculated using the formula**

$$\text{Return on Capital Employed} = \frac{\text{Operating Profit}}{\text{Capital Employed}} \times 100$$

Worked Example

The table shows an extract from the company accounts of *Keals Cosmetics*.

Current Liabilities	£1.5 million
Revenue	£7 million
Total Assets	£15.4 million
Operating Profit	£2.2 million

Calculate Keals Cosmetics' return on capital employed.

(3)

Step 1: Calculate the capital employed

$$= £15.4\text{m} - £1.5\text{m}$$ (1)

$$= £13.9\text{m}$$

Step 2: Divide the operating profit by the capital employed

$$= 0.16$$

Step 3: Multiply the result by 100, and express the outcome as a percentage

$$= 0.16 \times 100 \quad (1)$$

$$= 16\%$$

- The **capital employed** in Keals Cosmetics has **generated a return of 16%**

Interpreting ratios to make business decisions

Gearing ratio

- The gearing ratio is used to examine the **capital structure** of a business
- It compares the amount of **capital raised from shareholders** with **capital raised from loans** and other forms of long-term borrowing to show the **proportion of business assets that are financed by long-term borrowing**
- In short, it shows how **reliant** a business is upon **borrowed money**

Highly geared business

- In a **highly geared** business, **more than 50 per cent** of its capital employed are **long-term loans**
  - The outcome of the gearing ratio calculation will be greater than 50 per cent
  - Substantial levels of interest will need to be paid on this high level of borrowing, which means:
    - The level of profit available to pay as dividends to shareholders is reduced
    - Profit available to retain within the business is limited
    - The business is likely to be considered a risk for further investment
    - It is also likely to face difficulties in raising further loan capital

Steps to reduce gearing

- A highly geared business may take steps to lower its ratio by:
  - Issuing more ordinary shares to create further share capital
  - Retaining more profits to avoid further borrowing
  - Repaying loans to lower interest costs for the business

Low geared business

- A **low-geared** business has **less than 50 per cent** of its capital employed as **long-term loans**

to access finance without the need to dilute existing shareholders' control
    - This is especially true **when interest rates are very low**, as has been the case in the UK over the last 15 years
    - Lenders, such as banks, are **more likely to approve loan applications** from low-geared businesses
    - An **unwillingness to access loan capital** may indicate a **risk-averse** business, which **may deter investors**

Steps to increase gearing

- A low-geared business may take steps to increase its ratio by:
    - Buying back **ordinary shares** to reduce share capital in relation to borrowing
    - Issuing more preference shares with limited loss of control
    - **Obtaining more loans**

Worked Example

Catseye Pressings Ltd is considering making an application for a long-term loan to purchase a new storage facility.

The table shows extracts from its balance sheet.

Non-current Assets	£16.40m
Current Assets	£3.62m
Current Liabilities	£2.18m
Non-current Liabilities	£5.75m

Calculate *Catseye Pressings Ltd's* gearing ratio, and advise whether an application for a loan is likely to be approved on this basis.

(5)

Step 1: Calculate the capital employed

$$= (£16.40m + £3.62m) - £2.18m$$ (1)

$$= £17.84m$$

Step 2: Apply the formula to calculate gearing

$$= \frac{£5.75m}{£17.84m} \times 100$$ (1)

$$= 32.23\%$$

Step 3: Identify whether the loan application is likely to be approved

Interpreting return on capital employed (RoCE)

- RoCE measures how well a business **generates profit** from the **funds invested** in the business
- The rate **differs between industries**, so comparison across sectors is not recommended
- However, it can be compared with other forms of return, such as **interest rates on savings** and with other businesses within the **same industry**
- RoCE can be used to support **strategic decisions** (e.g. investment or divestment decisions) to determine the most profitable option given the level of capital employed
- With RoCE, **the higher the rate, the better,** as it indicates that the business is **profitable** and **using its capital efficiently**
  - Investors prefer businesses with **stable and rising** levels of RoCE, as this indicates **low-risk growth** is being achieved
  - A ROCE of **at least 20 per cent** is usually a good sign that the company is in a good financial position

- To increase the RoCE level, a business can:
  - **increase the level of profit** generated **without introducing new capital** into the business
  - **maintain the level of profit** generated **whilst reducing the amount of capital** in the business

Worked Example

Faced with increasing costs, *Kent & Medway Properties Ltd* is looking to close one of its three high street estate agency branches.

The table below shows some key data for each of the branches.

Branch	Capital Employed	Operating Profit
Sevenoaks	£2.4m	£0.37m
Whitstable	£3.1m	£0.57m
Rochester	£2.9m	£0.51m

Calculate the return on capital employed (RoCE) for each branch, and recommend which branch, on profitability terms, should close.

(5)

Step 1: Apply the formula to calculate the RoCE for each branch

$$\text{Sevenoaks} = \frac{\text{£}2.4\text{m}}{} \times 100 = 15.42\%$$

$$\text{Whitstable} = \frac{\text{£}0.57\text{m}}{\text{£}3.1\text{m}} \times 100 = 18.39\%$$

$$\text{Rochester} = \frac{\text{£}0.51\text{m}}{\text{£}2.9\text{m}} \times 100 = 17.59\%$$

(3)

Step 2: Identify the least profitable branch for closure

- Sevenoaks is the least profitable branch, **(1)** with a RoCE of 15.42%, **(1)** and should be the branch selected for closure

Examiner Tips and Tricks

When calculating financial ratios, check that you are using the correct units.

In some cases, financial data is presented as raw figures (e.g. £14,520), but in most cases, you will be working in thousands (£000) or millions (£m).

- Ensure that you convert correctly, e.g. £0.39m is equal to £390,000, and £34.9 (000) is equal to £34,900
- Make sure the decimal is in the correct place
- Calculate to two decimal places unless stated otherwise

The limitations of ratio analysis

- There are some drawbacks of using ratio analysis, which businesses need to be aware of

Limitations of ratio analysis

Limitation	Explanation

Making comparisons	• It is important to compare **like for like**
	• **Over time**, the **nature of a business** can change, affecting the desired level of ratio
	  ○ E.g. A business may diversify into a less competitive industry where higher levels of RoCE may be expected
	• Comparisons **between firms are** only meaningful where significant similarities exist (e.g. same industry, similar size, comparable products)
	  ○ E.g. a high street jewellery business will have very different

Quality of accounts	• Accounts may have been legally **window dressed (manipulated)** to present a particular financial picture.
	• Examples of this include:
	  ○ Bad debts can be written off
	  ○ Property can be revalued
	  ○ Income and costs may be reported during an earlier or later reporting period (e.g. payments to suppliers may be delayed to maximise the level of current assets)
	• Window dressing will have an impact on the **quality** of ratio analysis calculations

Statement of financial position limitations	• As a "**snapshot**" of a business's assets, liabilities and capital at a specific point of time, the **balance sheet may not be representative** of its usual circumstances ○ E.g. a balance sheet may be completed one day before a business sells a large amount of stock or buys a new property, rendering current and non-current assets figures invalid almost immediately
Qualitative information	• As ratios **only use numerical data** from a business's accounts, key qualitative factors that affect its performance are ignored ○ E.g. the collapse of a competitor may lead to increased sales revenue and profit • Increased profit increases the RoCE without any strategic decisions being made

- Despite these limitations, ratio analysis is **used by a wide range of internal and external analysts** to assess the performance of companies
  - **Venture capitalists** and other investors use ratios to support their analysis when they consider **investing in** or **lending to** businesses
  - **Banks** and **insurance providers** will use ratios to determine the **level of risk** a business presents and determine the products to which it may be suited
  - **Investment analysts** and **journalists** make use of ratio analysis to **report to clients and the media** in easy-to-digest terms

Examiner Tips and Tricks

Calculating ratios is straightforward — the real skill is in the interpretation of results and making recommendations.

If a business is highly geared, further borrowing is likely to be neither attractive nor possible in most cases. That's quite a simple analytical point.

However, it may be possible to further develop this analysis.

For example, in a time of very low interest rates, maximising borrowing to take advantage of cheap finance may be preferable to diluting the control of existing shareholders by issuing further share capital. Now you have an evaluation because you've considered both sides of the argument.

If you find evidence in the case study that indicates that shareholders would be unhappy with this dilution of control, then you have a balanced, applied piece of evaluation.

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